RETAIL SALES DATA CLEANING AND QUALITY ASSESSMENT

Excel Portfolio Project

Dataset: 200 Fictional Retail Sales Records

Tool Used: Microsoft Excel

Project Focus: Data Cleaning | Data Quality Checks | Data Validation | Reporting

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Project Type: Independent Practice Project

Purpose: To demonstrate practical Excel skills in reviewing, cleaning, organizing, and assessing data quality.

1. PROJECT OVERVIEW

Project Background

This project involved cleaning and reviewing a dataset containing 200 fictional retail sales records using Microsoft Excel.

The purpose was to identify data-quality problems, organize the information, standardize entries where appropriate, and document records requiring further review.

Project Objectives

- Identify missing and duplicate records.
- Detect inconsistent product information and formatting.
- Review invalid entries and incorrect sales totals.
- Standardize information where reliable corrections could be made.
- Document unresolved issues and recommend appropriate actions.

Tools and Techniques

Software: Microsoft Excel

Techniques used:

- Excel formulas and functions.
- Sorting and filtering.
- Text and date standardization.
- Duplicate and missing-value checks.
- Data validation and cross-checking.
- Quality reporting and documentation.

Dataset Information

Item	Details
Dataset	Retail Sales
Records	200
Data type	Fictional
Software	Microsoft Excel
Project type	Independent practice project

2. WORKBOOK STRUCTURE AND CLEANING PROCESS

Workbook Structure

The Excel workbook contains seven worksheets, each serving a specific purpose:

Worksheet	Purpose
Raw_Data	Preserves the original dataset unchanged.
Working	Provides space for reviewing and cleaning records.
Cleaned_Data	Contains the reviewed and standardized dataset.
Quality_Report	Documents individual findings and recommended actions.
Quality_Summary	Summarizes the 11 data-quality issue categories.
Input_Review	Documents input-related issues.
Total_Review	Records sales total checks and mismatches.

Cleaning Process

I followed a structured process:

1. Reviewed the original dataset without changing it.
2. Checked for missing values, duplicates, and inconsistent information.
3. Used Excel formulas and other checks to investigate possible errors.
4. Standardized information where the correct values could be established.
5. Compared records and reviewed sales totals.
6. Documented findings and recommended actions.
7. Preserved unresolved records for further verification.

Data Handling Approach

The original dataset was preserved throughout the project. I avoided making assumptions when information could not be confirmed and documented unresolved issues for further review.

3. DATA QUALITY FINDINGS AND RESULTS

Quality Assessment Summary

During the review, I documented 24 individual findings across 11 data-quality issue categories.

The findings covered areas such as:

- Missing information.
- Duplicate records.
- Inconsistent product information.
- Formatting issues.
- Invalid input values.
- Incorrect sales totals.
- Records requiring further verification.

Review and Verification

I used Excel formulas, comparisons, and validation checks to investigate the identified issues.

Each finding was documented with its affected records, a description of the issue, a recommended action, and a verification status.

Handling Unresolved Issues

Not every issue could be corrected using the available information. Where the correct value could not be confirmed, I left the record unchanged and marked it for further verification.

This helped preserve the original information and avoid introducing unsupported corrections.

Project Results

The project produced:

- A preserved copy of the original dataset.
- A separate cleaned and standardized dataset.
- A quality summary covering 11 issue categories.
- Findings report containing 24 documented findings.
- Supporting reviews of input-related issues and sales total mismatches.

4. PROJECT EVIDENCE AND CONCLUSION

Project Evidence

The following screenshots provide visual evidence of the work completed:

1. Workbook overview — shows the organization of the seven worksheets.
2. Working Sheet — shows the workspace used to review records, apply cleaning steps, and investigate data-quality issues.
3. Quality summary — presents the 11 data-quality issue categories.
4. Quality report — shows the 24 documented findings and recommended actions.
5. Cleaned data — displays the reviewed and standardized dataset.
6. Input review — shows the input-related checks.
7. Total review — presents the sales total checks.

Conclusion

This project provided practical experience in using Microsoft Excel to review, clean, organize, and assess data quality.

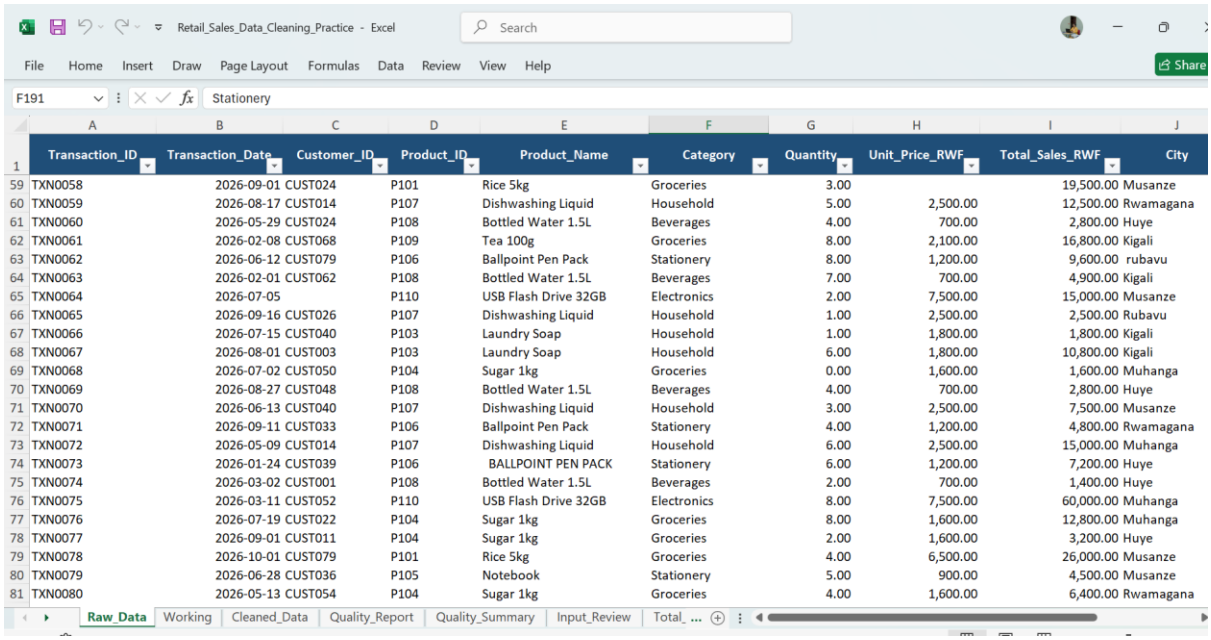
It strengthened my skills in spreadsheet management, data cleaning, data validation, record checking, and reporting.

The project also reinforced the importance of preserving original information and verifying uncertain records before making corrections.

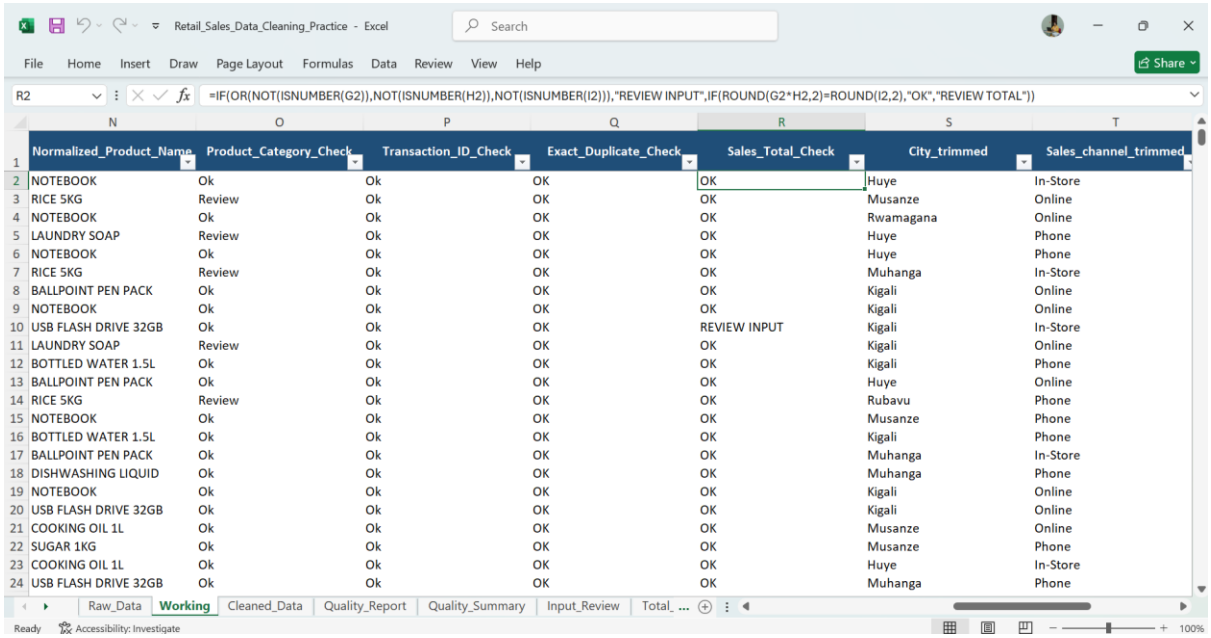
Project Status: Completed as an independent practice project. Some findings remain pending verification.

Dataset: 200 fictional retail sales records.

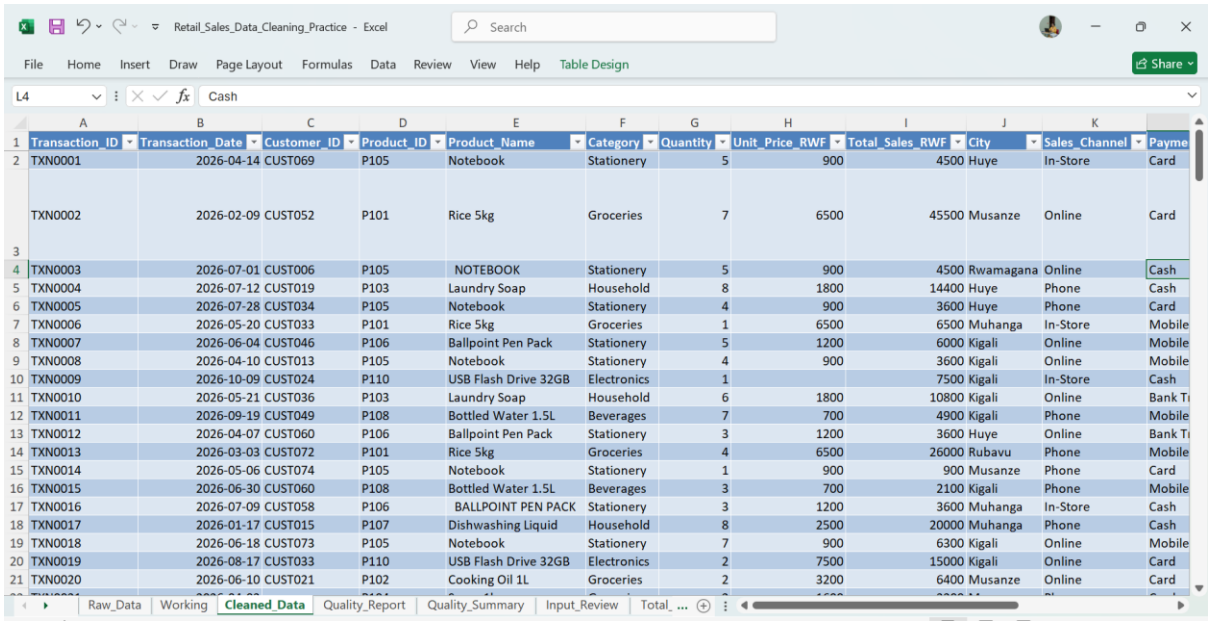
Software: Microsoft Excel



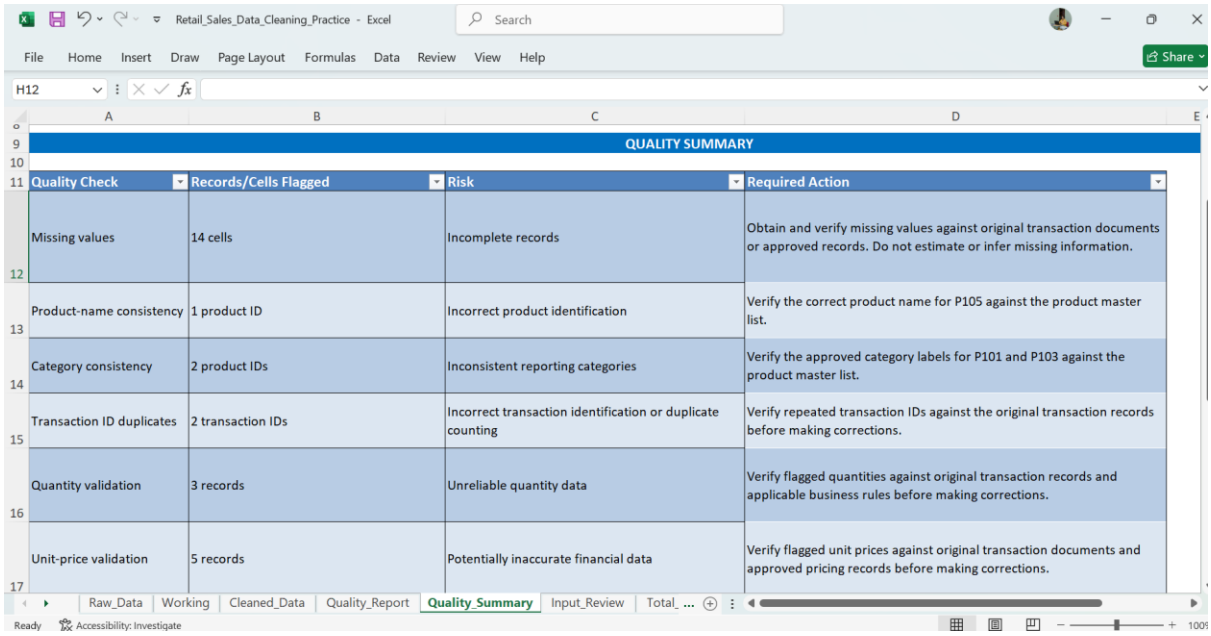
Transaction_ID	Transaction_Date	Customer_ID	Product_ID	Product_Name	Category	Quantity	Unit_Price_RWF	Total_Sales_RWF	City
TXN0058	2026-09-01	CUST024	P101	Rice 5kg	Groceries	3.00		19,500.00	Musanze
TXN0059	2026-08-17	CUST014	P107	Dishwashing Liquid	Household	5.00	2,500.00	12,500.00	Rwamagana
TXN0060	2026-05-29	CUST024	P108	Bottled Water 1.5L	Beverages	4.00	700.00	2,800.00	Huye
TXN0061	2026-02-08	CUST068	P109	Tea 100g	Groceries	8.00	2,100.00	16,800.00	Kigali
TXN0062	2026-06-12	CUST079	P106	Ballpoint Pen Pack	Stationery	8.00	1,200.00	9,600.00	rubavu
TXN0063	2026-02-01	CUST062	P108	Bottled Water 1.5L	Beverages	7.00	700.00	4,900.00	Kigali
TXN0064	2026-07-05		P110	USB Flash Drive 32GB	Electronics	2.00	7,500.00	15,000.00	Musanze
TXN0065	2026-09-16	CUST026	P107	Dishwashing Liquid	Household	1.00	2,500.00	2,500.00	Rubavu
TXN0066	2026-07-15	CUST040	P103	Laundry Soap	Household	1.00	1,800.00	1,800.00	Kigali
TXN0067	2026-08-01	CUST003	P103	Laundry Soap	Household	6.00	1,800.00	10,800.00	Kigali
TXN0068	2026-07-02	CUST050	P104	Sugar 1kg	Groceries	0.00	1,600.00	1,600.00	Muhanga
TXN0069	2026-08-27	CUST048	P108	Bottled Water 1.5L	Beverages	4.00	700.00	2,800.00	Huye
TXN0070	2026-06-13	CUST040	P107	Dishwashing Liquid	Household	3.00	2,500.00	7,500.00	Musanze
TXN0071	2026-09-11	CUST033	P106	Ballpoint Pen Pack	Stationery	4.00	1,200.00	4,800.00	Rwamagana
TXN0072	2026-05-09	CUST014	P107	Dishwashing Liquid	Household	6.00	2,500.00	15,000.00	Muhanga
TXN0073	2026-01-24	CUST039	P106	BALLPOINT PEN PACK	Stationery	6.00	1,200.00	7,200.00	Huye
TXN0074	2026-03-02	CUST001	P108	Bottled Water 1.5L	Beverages	2.00	700.00	1,400.00	Huye
TXN0075	2026-03-11	CUST052	P110	USB Flash Drive 32GB	Electronics	8.00	7,500.00	60,000.00	Muhanga
TXN0076	2026-07-19	CUST022	P104	Sugar 1kg	Groceries	8.00	1,600.00	12,800.00	Muhanga
TXN0077	2026-09-01	CUST011	P104	Sugar 1kg	Groceries	2.00	1,600.00	3,200.00	Huye
TXN0078	2026-10-01	CUST079	P101	Rice 5kg	Groceries	4.00	6,500.00	26,000.00	Musanze
TXN0079	2026-06-28	CUST036	P105	Notebook	Stationery	5.00	900.00	4,500.00	Musanze
TXN0080	2026-05-13	CUST054	P104	Sugar 1kg	Groceries	4.00	1,600.00	6,400.00	Rwamagana



Normalized_Product_Name	Product_Category_Check	Transaction_ID_Check	Exact_Duplicate_Check	Sales_Total_Check	City_trimmed	Sales_channel_trimmed
NOTEBOOK	Ok	Ok	OK	OK	Huye	In-Store
RICE 5KG	Review	Ok	OK	OK	Musanze	Online
NOTEBOOK	Ok	Ok	OK	OK	Rwamagana	Online
LAUNDRY SOAP	Review	Ok	OK	OK	Huye	Phone
NOTEBOOK	Ok	Ok	OK	OK	Huye	Phone
RICE 5KG	Review	Ok	OK	OK	Muhanga	In-Store
BALLPOINT PEN PACK	Ok	Ok	OK	OK	Kigali	Online
NOTEBOOK	Ok	Ok	OK	OK	Kigali	Online
USB FLASH DRIVE 32GB	Ok	Ok	OK	REVIEW INPUT	Kigali	In-Store
LAUNDRY SOAP	Review	Ok	OK	OK	Kigali	Online
BOTTLED WATER 1.5L	Ok	Ok	OK	OK	Kigali	Phone
BALLPOINT PEN PACK	Ok	Ok	OK	OK	Huye	Online
RICE 5KG	Review	Ok	OK	OK	Rubavu	Phone
NOTEBOOK	Ok	Ok	OK	OK	Musanze	Phone
BOTTLED WATER 1.5L	Ok	Ok	OK	OK	Kigali	Phone
BALLPOINT PEN PACK	Ok	Ok	OK	OK	Muhanga	In-Store
DISHWASHING LIQUID	Ok	Ok	OK	OK	Muhanga	Phone
NOTEBOOK	Ok	Ok	OK	OK	Kigali	Online
USB FLASH DRIVE 32GB	Ok	Ok	OK	OK	Kigali	Online
COOKING OIL 1L	Ok	Ok	OK	OK	Musanze	Online
SUGAR 1KG	Ok	Ok	OK	OK	Musanze	Phone
COOKING OIL 1L	Ok	Ok	OK	OK	Huye	In-Store
USB FLASH DRIVE 32GB	Ok	Ok	OK	OK	Muhanga	Phone

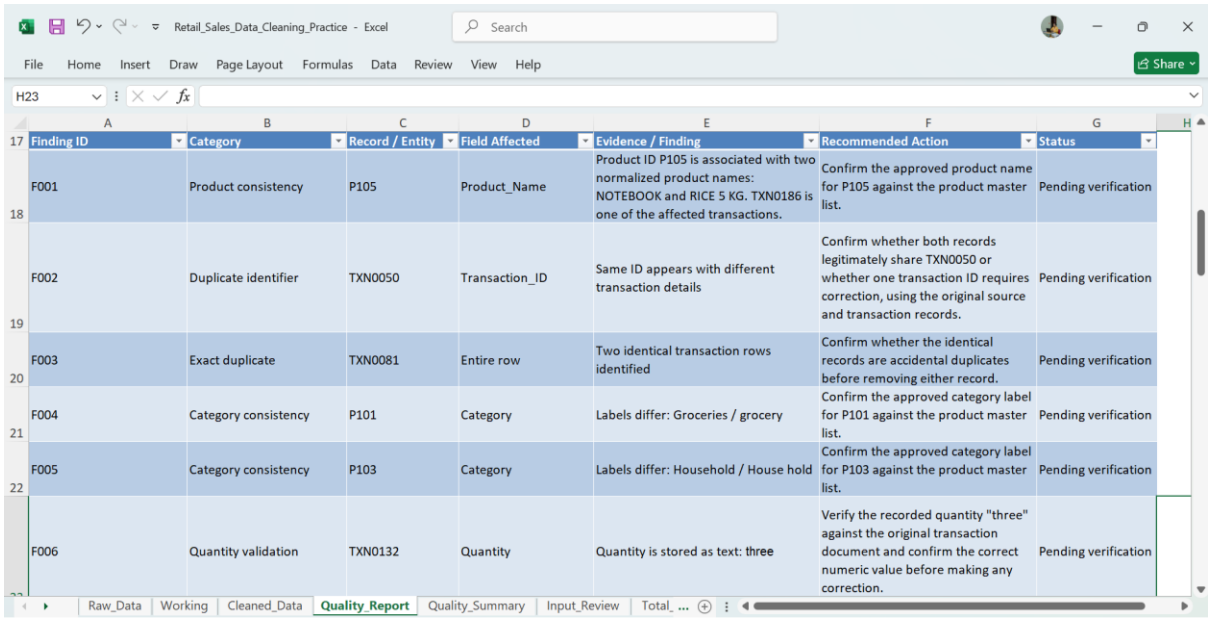


Transaction_ID	Transaction_Date	Customer_ID	Product_ID	Product_Name	Category	Quantity	Unit_Price_RWF	Total_Sales_RWF	City	Sales_Channel	Payme
TXN0001	2026-04-14	CUST069	P105	Notebook	Stationery	5	900	4500	Huye	In-Store	Card
TXN0002	2026-02-09	CUST052	P101	Rice 5kg	Groceries	7	6500	45500	Musanze	Online	Card
TXN0003	2026-07-01	CUST006	P105	NOTEBOOK	Stationery	5	900	4500	Rwamagana	Online	Cash
TXN0004	2026-07-12	CUST019	P103	Laundry Soap	Household	8	1800	14400	Huye	Phone	Cash
TXN0005	2026-07-28	CUST034	P105	Notebook	Stationery	4	900	3600	Huye	Phone	Card
TXN0006	2026-05-20	CUST033	P101	Rice 5kg	Groceries	1	6500	6500	Muhanga	In-Store	Mobile
TXN0007	2026-06-04	CUST046	P106	Ballpoint Pen Pack	Stationery	5	1200	6000	Kigali	Online	Mobile
TXN0008	2026-04-10	CUST013	P105	Notebook	Stationery	4	900	3600	Kigali	Online	Mobile
TXN0009	2026-10-09	CUST024	P110	USB Flash Drive 32GB	Electronics	1		7500	Kigali	In-Store	Cash
TXN0010	2026-05-21	CUST036	P103	Laundry Soap	Household	6	1800	10800	Kigali	Online	Bank T
TXN0011	2026-09-19	CUST049	P108	Bottled Water 1.5L	Beverages	7	700	4900	Kigali	Phone	Mobile
TXN0012	2026-04-07	CUST060	P106	Ballpoint Pen Pack	Stationery	3	1200	3600	Huye	Online	Bank T
TXN0013	2026-03-03	CUST072	P101	Rice 5kg	Groceries	4	6500	26000	Rubavu	Phone	Mobile
TXN0014	2026-05-06	CUST074	P105	Notebook	Stationery	1	900	900	Musanze	Phone	Card
TXN0015	2026-06-30	CUST060	P108	Bottled Water 1.5L	Beverages	3	700	2100	Kigali	Phone	Mobile
TXN0016	2026-07-09	CUST058	P106	BALLPOINT PEN PACK	Stationery	3	1200	3600	Muhanga	In-Store	Cash
TXN0017	2026-01-17	CUST015	P107	Dishwashing Liquid	Household	8	2500	20000	Muhanga	Phone	Cash
TXN0018	2026-06-18	CUST073	P105	Notebook	Stationery	7	900	6300	Kigali	Online	Mobile
TXN0019	2026-08-17	CUST033	P110	USB Flash Drive 32GB	Electronics	2	7500	15000	Kigali	Online	Card
TXN0020	2026-06-10	CUST021	P102	Cooking Oil 1L	Groceries	2	3200	6400	Musanze	Online	Card

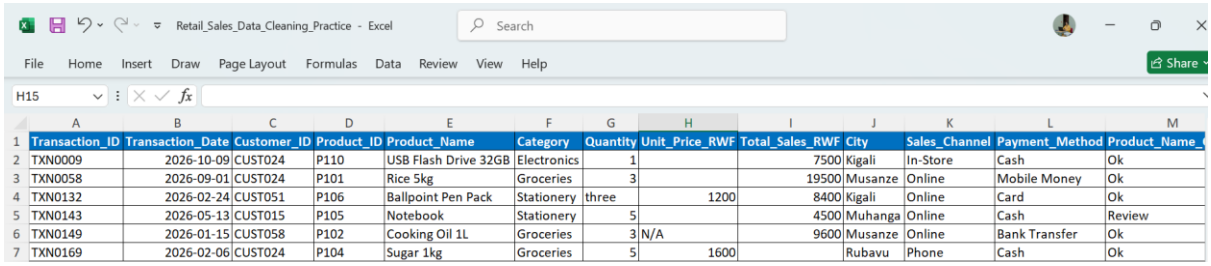


QUALITY SUMMARY

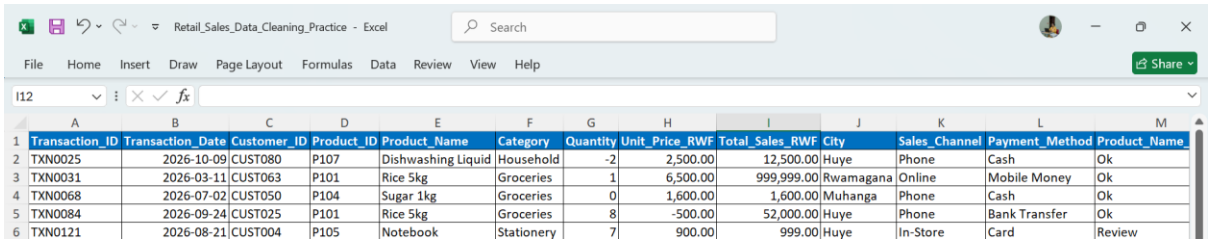
Quality Check	Records/Cells Flagged	Risk	Required Action
Missing values	14 cells	Incomplete records	Obtain and verify missing values against original transaction documents or approved records. Do not estimate or infer missing information.
Product-name consistency	1 product ID	Incorrect product identification	Verify the correct product name for P105 against the product master list.
Category consistency	2 product IDs	Inconsistent reporting categories	Verify the approved category labels for P101 and P103 against the product master list.
Transaction ID duplicates	2 transaction IDs	Incorrect transaction identification or duplicate counting	Verify repeated transaction IDs against the original transaction records before making corrections.
Quantity validation	3 records	Unreliable quantity data	Verify flagged quantities against original transaction records and applicable business rules before making corrections.
Unit-price validation	5 records	Potentially inaccurate financial data	Verify flagged unit prices against original transaction documents and approved pricing records before making corrections.



Finding ID	Category	Record / Entity	Field Affected	Evidence / Finding	Recommended Action	Status
F001	Product consistency	P105	Product_Name	Product ID P105 is associated with two normalized product names: NOTEBOOK and RICE 5 KG. TXN0186 is one of the affected transactions.	Confirm the approved product name for P105 against the product master list.	Pending verification
F002	Duplicate identifier	TXN0050	Transaction_ID	Same ID appears with different transaction details	Confirm whether both records legitimately share TXN0050 or whether one transaction ID requires correction, using the original source and transaction records.	Pending verification
F003	Exact duplicate	TXN0081	Entire row	Two identical transaction rows identified	Confirm whether the identical records are accidental duplicates before removing either record.	Pending verification
F004	Category consistency	P101	Category	Labels differ: Groceries / grocery	Confirm the approved category label for P101 against the product master list.	Pending verification
F005	Category consistency	P103	Category	Labels differ: Household / House hold	Confirm the approved category label for P103 against the product master list.	Pending verification
F006	Quantity validation	TXN0132	Quantity	Quantity is stored as text: three	Verify the recorded quantity "three" against the original transaction document and confirm the correct numeric value before making any correction.	Pending verification



Transaction_ID	Transaction_Date	Customer_ID	Product_ID	Product_Name	Category	Quantity	Unit_Price_RWF	Total_Sales_RWF	City	Sales_Channel	Payment_Method	Product_Name_
TXN0009	2026-10-09	CUST024	P110	USB Flash Drive 32GB	Electronics	1		7500	Kigali	In-Store	Cash	Ok
TXN0058	2026-09-01	CUST024	P101	Rice 5kg	Groceries	3		19500	Musanze	Online	Mobile Money	Ok
TXN0132	2026-02-24	CUST051	P106	Ballpoint Pen Pack	Stationery	three	1200	8400	Kigali	Online	Card	Ok
TXN0143	2026-05-13	CUST015	P105	Notebook	Stationery	5		4500	Muhanga	Online	Cash	Review
TXN0149	2026-01-15	CUST058	P102	Cooking Oil 1L	Groceries	3	N/A	9600	Musanze	Online	Bank Transfer	Ok
TXN0169	2026-02-06	CUST024	P104	Sugar 1kg	Groceries	5	1600		Rubavu	Phone	Cash	Ok



Transaction_ID	Transaction_Date	Customer_ID	Product_ID	Product_Name	Category	Quantity	Unit_Price_RWF	Total_Sales_RWF	City	Sales_Channel	Payment_Method	Product_Name
TXN0025	2026-10-09	CUST080	P107	Dishwashing Liquid	Household	-2	2,500.00	12,500.00	Huye	Phone	Cash	Ok
TXN0031	2026-03-11	CUST063	P101	Rice 5kg	Groceries	1	6,500.00	999,999.00	Rwamagana	Online	Mobile Money	Ok
TXN0068	2026-07-02	CUST050	P104	Sugar 1kg	Groceries	0	1,600.00	1,600.00	Muhanga	Phone	Cash	Ok
TXN0084	2026-09-24	CUST025	P101	Rice 5kg	Groceries	8	-500.00	52,000.00	Huye	Phone	Bank Transfer	Ok
TXN0121	2026-08-21	CUST004	P105	Notebook	Stationery	7	900.00	999.00	Huye	In-Store	Card	Review